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Department of CSE (Artificial Intelligence & Machine Learning)

ROBUST AUDIO-VISUAL SPEECH SEPARATION SYSTEM

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INTRODUCTION

Speech separation helps pick out one person's voice from a recording where several people are talking together.

In real-world environments such as meetings, classrooms, and video conferencing platforms, multiple people often speak simultaneously. Traditional audio-only separation systems struggle in noisy and overlapping speech conditions.

Recent research in deep learning show that visual cues such as lip motion can improve speech separation performance.

This project focuses on designing a robust audio-visual speech separation system that performs effectively even when video input suffers from low frame rate or dropped frames.

PROBLEM STATEMENT

In real-world video calls and recordings:

- Multiple speakers talk simultaneously
- Background noise corrupts audio
- Video frames may be dropped or delayed
- Visual input can be inconsistent

Existing audio-visual models assume high-quality synchronized video and fail under low-frame-rate conditions.

Therefore, there is a need to develop a robust audio-visual speech separation system that can dynamically adapt when video input is unreliable.

OBJECTIVES

- To study existing audio-visual speech separation techniques
- To develop a multimodal deep learning model combining audio and visual features
- To design a temporal attention mechanism for handling low-frame-rate video
- To allow user-selected speaker extraction
- To evaluate performance under corrupted and low-FPS conditions

MOTIVATION

- Increasing use of video conferencing platforms
- Real-world video often suffers from lag and frame drops
- Existing systems work well only in controlled environments
- Practical need for robust speech separation
- Improve accessibility and speech clarity in noisy environments

LITERATURE SURVEY

Paper / Author	Key Methodology	Limitation
Looking to Listen *(Ephrat et al., 2018)* *(Base Paper)*	Uses a two-stream CNN to fuse audio and extracted face/lip features for separation.	**Assumes stable video quality.** Performance drops significantly if frames are missing or video lags.
The Conversation *(Afouras et al., 2018)*	Enhances both magnitude and phase of speech using deep audio-visual features.	**Computationally expensive.** Does not account for variable frame rates common in real-world calls.
Modality Attention *(Ma et al., 2019)*	Introduces an attention mechanism to weigh audio vs. video importance dynamically.	Focused on **Speech Recognition (ASR)**, not signal reconstruction. Struggles when video is reliable but low-FPS.

NOVELTY

EXISTING SYSTEM

- Uses only audio to separate voices
- Some systems use both audio and video
- Assumes video is clear and synchronized
- Performance drops when video frames are missing
- Cannot adapt to low frame rate
- Not fully suitable for real-world video calls

PROPOSED SYSTEM

- Uses audio and visual (lip motion) features for speech separation
- Introduces temporal attention for dynamic audio-visual fusion
- Adapts automatically when visual frames are unreliable
- Maintains performance under low frame rate and dropped frames
- Allows target speaker selection from multi-speaker video
- Generates clean separated speech output

SCOPE

- Application in video conferencing platforms
- Assistive technology for hearing-impaired individuals
- Speech enhancement in surveillance systems
- Real-world noisy communication environments

CHALLENGES

- Synchronization of audio and video streams
- Handling different frame rates
- Training the model
- Dataset preprocessing

PO'S, PSO'S AND SDG'S

PO's:

- Engineering knowledge & Problem analysis (PO1, PO2)
- Design and development of solutions using modern tools (PO3, PO5)

PSO's:

- Application of AI & ML in real-world systems (PSO1)
- Development of intelligent multimedia systems (PSO2)

SDG:

- SDG 9: Industry, Innovation and Infrastructure
- SDG 4: Quality Education (assistive communication systems)

REFERENCES

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THANK YOU